VoxHub CM4 — Home Assistant voice hub
USB-C PD (9–20 V) -> 5 V buck -> CM4 Lite + USB hub + audio + Zigbee

Power: USB-C PD sink, 5 V buck

File: power.kicad_sch

Compute Module 4 socket, microSD, user interface

File: cm4.kicad_sch

USB 2.0 hub and USB-A host ports

File: usb.kicad_sch

Audio: PDM microphones, PDM-to-I2S, class-D speaker amplifier

File: audio.kicad_sch

Zigbee / Thread radio (MGM240P, EFR32MG24)

File: zigbee.kicad_sch

H1
M2.5 board mount

H2
M2.5 board mount

H3
M2.5 board mount

H4
M2.5 board mount

FID1
Fiducial

FID2
Fiducial

FID3
Fiducial

H5
CM4 spacer M2.5x1.5

GND

H6
CM4 spacer M2.5x1.5

GND

H7
CM4 spacer M2.5x1.5

GND

H8
CM4 spacer M2.5x1.5

GND

Generated by hardware/gen — do not edit by hand

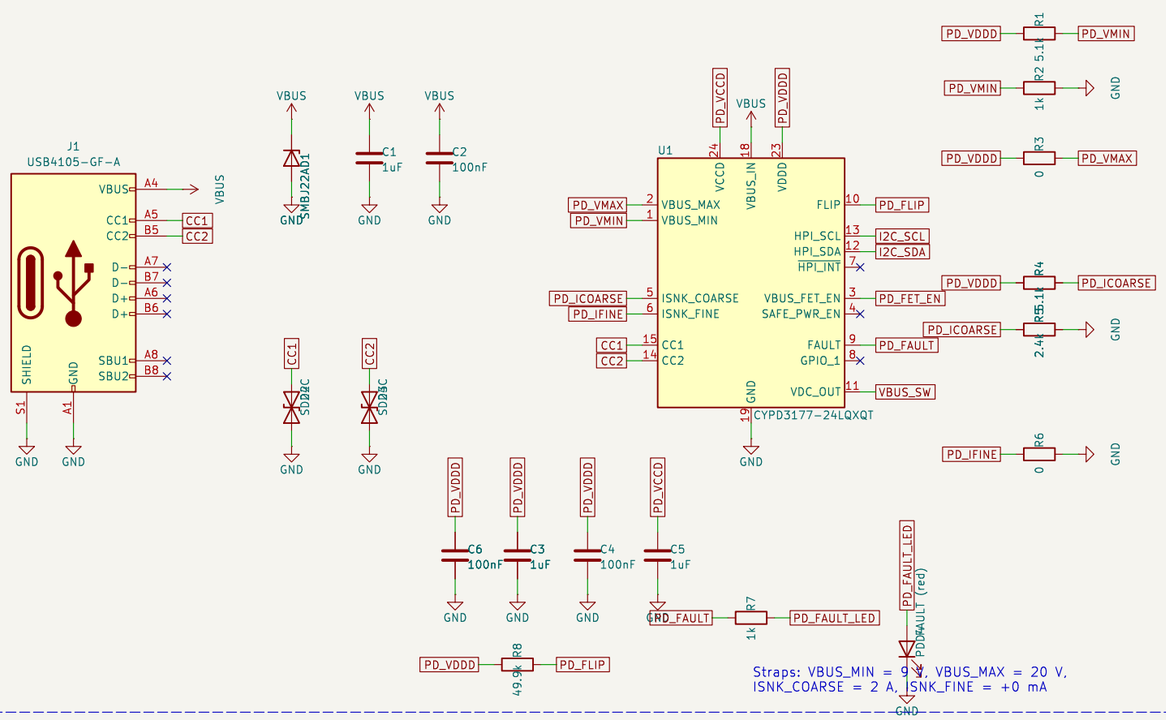
VoxHub (open hardware)

Sheet: /
File: voxhub.kicad_sch

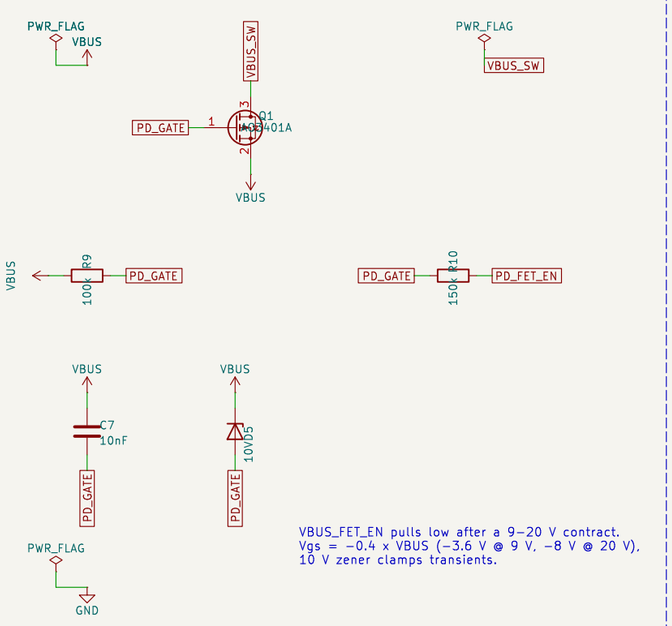
Title: VoxHub CM4 — Home Assistant voice hub carrier

Size: A3	Date: 2026-11-11	**Rev: A1**
KiCad E.D.A. 9.0.9		Id: 1/6

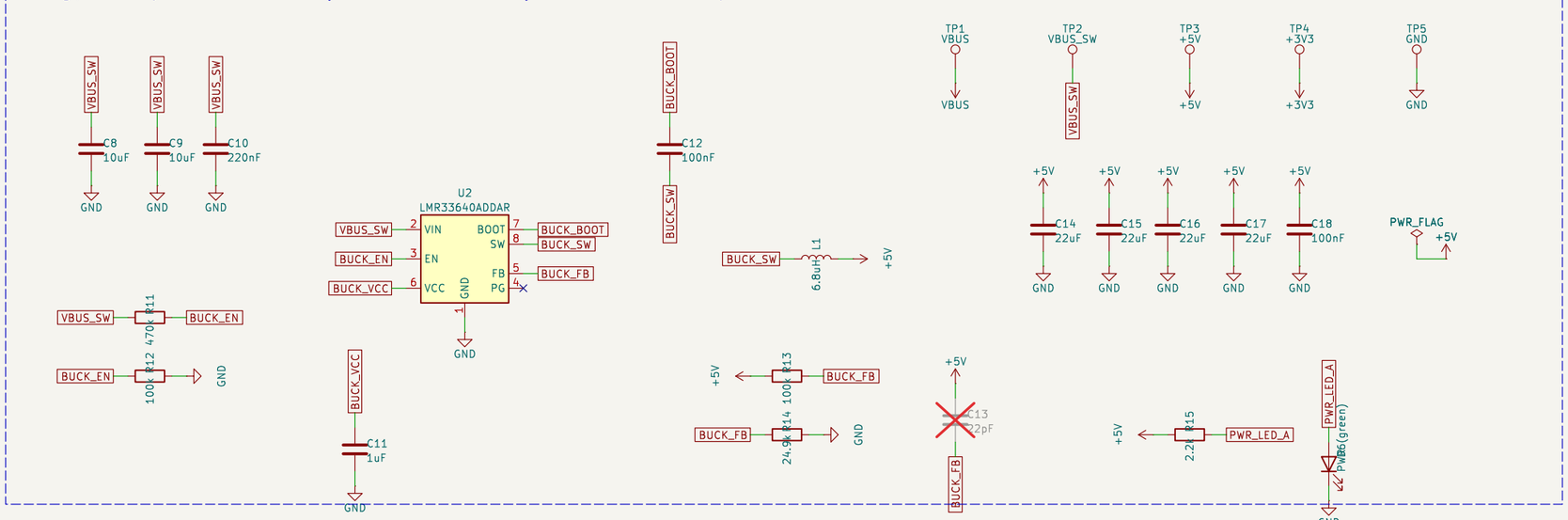
USB-C input and PD sink controller (CYPD3177 'BCR', resistor-strapped: 9–20 V, 2 A minimum)



VBUS load switch: P-FET driven by the BCR's active-low, slew-controlled gate drive



5 V synchronous buck (LMR33640, 400 kHz) — UVLO 7.0 V keeps it off at vSafe5V

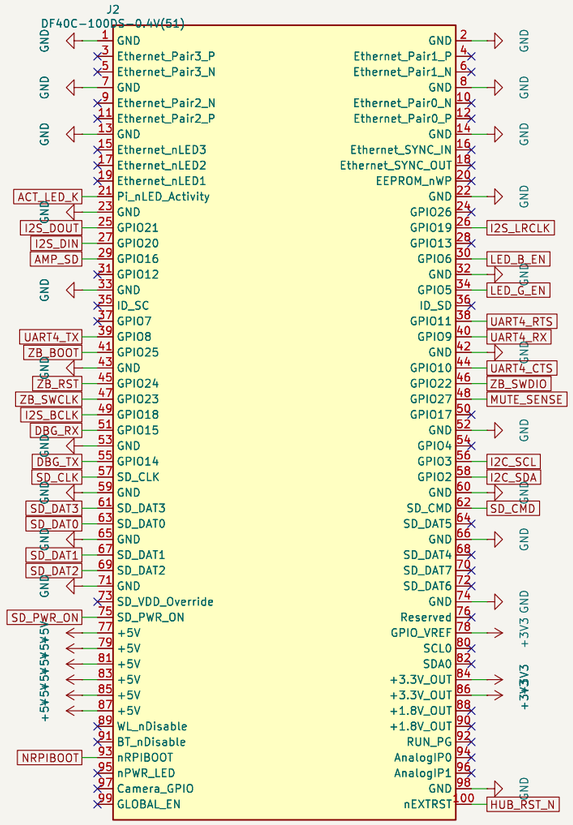


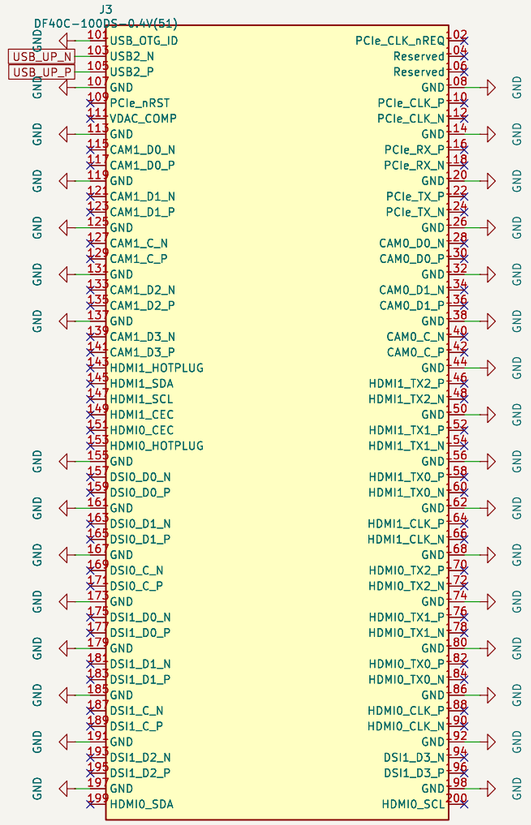
Generated by hardware/gen — do not edit by hand
VoxHub (open hardware)

Sheet: /Power: USB-C PD sink, 5 V buck/
File: power.kicad_sch

Title: Power: USB-C PD sink, 5 V buck

Size: A3	Date: 2026-11-11	**Rev: A1**
KiCad E.D.A. 9.0.9		Id: 2/6

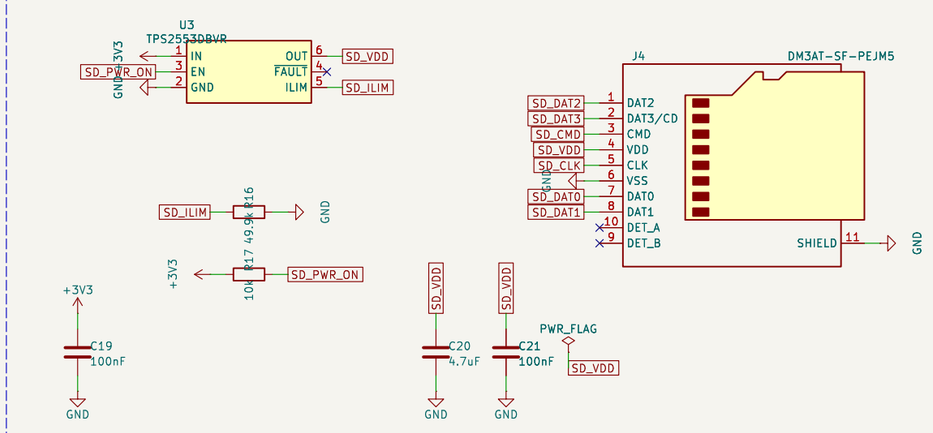




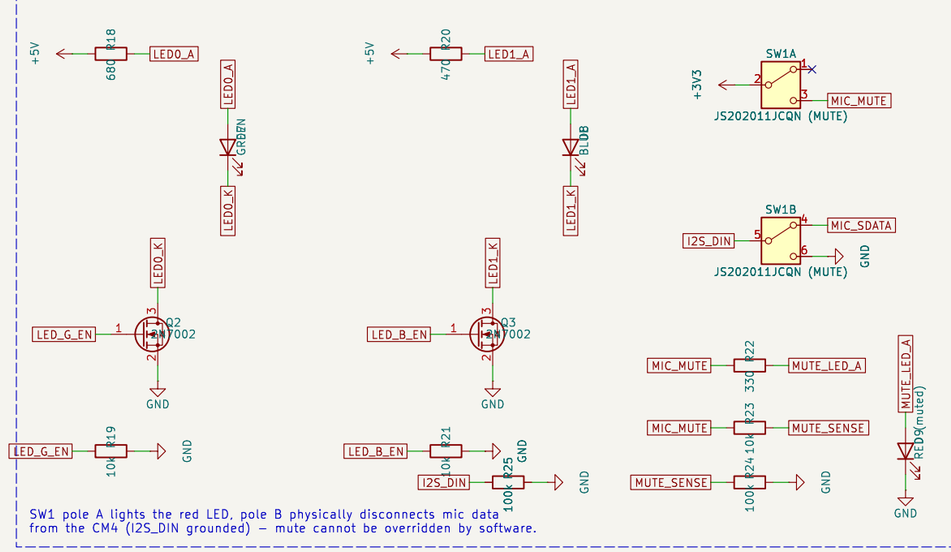


Module: Raspberry Pi CM4 Lite, wireless (CM4104000 / CM4102000 / CM4108000)
GPIO map: see README. USB_OTG_ID tied low = host.

microSD (boot / storage) — card power switched by SD_PWR_ON

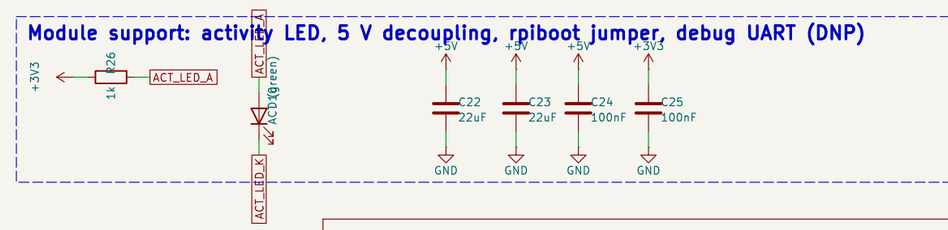


User interface: status LEDs (reverse-mount, shine through PCB to the top cover) and hardware mi



SW1 pole A lights the red LED, pole B physically disconnects mic data from the CM4 (I2S_DIN grounded) — mute cannot be overridden by software.

Module support: activity LED, 5 V decoupling, rpiboot jumper, debug UART (DNP)



Generated by hardware/gen — do not edit by hand
VoxHub (open hardware)

Sheet: /Compute Module 4 socket, microSD, user interface/		
File: cm4.kicad_sch		
Title: Compute Module 4 socket, microSD, user interface		
Size: A3	Date: 2026-11-11	**Rev: A1**
KiCad E.D.A. 9.0.9		Id: 3/6

USB2514B hub: upstream = CM4 USB2, ports 1–2 = USB-A, ports 3–4 disabled by strap

U4
USB2514B-I/M2

RESET_N is driven by the CM4 (nEXTRST, pin 100); low during reset, high once the SoC boots.

VBUS_DET
USBDM_UP
USBDP_UP
RESET_N
TEST
RBIAS
XTALIN/CLKIN
XTALOUT
PLLFILT
CRFILT
SCL/SMBCLK/CFG_SEL0
SDA/SMBDATA/NON_REM1
SUSP_IND/LOCAL_PWR/NON_REM0
HS_IND/CFG_SEL1
VDDA33
VDD33
VSS
USBDM_DN1/PRT_DIS_M1
USBDP_DN1/PRT_DIS_P1
PRTPWR1/BC_EN1
OCS_N1
USBDM_DN2/PRT_DIS_M2
USBDP_DN2/PRT_DIS_P2
PRTPWR2/BC_EN2
OCS_N2
USBDM_DN3/PRT_DIS_M3
USBDP_DN3/PRT_DIS_P3
PRTPWR3/BC_EN3
OCS_N3
USBDM_DN4/PRT_DIS_M4
USBDP_DN4/PRT_DIS_P4
PRTPWR4/BC_EN4
OCS_N4

HUB_RBIAS R27 12k GND
HUB_PLLFILT C28 100nF GND
HUB_CRFILT C29 100nF GND
HUB_VBUSDET R28 100k +3V3
HUB_CFG0 R29 100k GND
HUB_CFG1 R30 100k GND
HUB_NONREM0 R31 100k GND
HUB_NONREM1 R32 100k GND
HUB_DIS3M R33 10k +3V3
HUB_DIS3P R34 10k +3V3
HUB_DIS4M R35 10k +3V3
HUB_DIS4P R36 10k +3V3
Y1 24MHz HUB_XI HUB_XO
C26 27pF HUB_XI GND
C27 27pF HUB_XO GND
C30 100nF +3V3 GND
C31 100nF +3V3 GND
C32 100nF +3V3 GND
C33 100nF +3V3 GND
C34 100nF +3V3 GND
C35 100nF +3V3 GND
C36 4.7uF +3V3 GND

Downstream ports: per-port 1.2 A current-limited switch, ESD protection, stacked USB-A

U5
TPS2553DBVR
IN
EN
GND
OUT
FAULT
ILIM
USB_PWR1 +5V GND
USB1_VBUS
USB_OC1_N
USB1_ILIM
R37 20k USB1_ILIM GND
R38 10k +3V3 USB_OC1_N
C37 4.7uF +5V GND

U7
USBLC6-2SC6
USB1_N
USB1_P
USB1_VBUS GND
C38 100uF USB1_VBUS GND
C39 100nF USB1_VBUS GND
PWR_FLAG USB1_VBUS

J6
61400826021
VBUS1
VBUS2
D1+
D1-
D2+
D2-
Shield
GND1
GND2
USB1_VBUS
USB2_VBUS
USB1_P
USB1_N
USB2_P
USB2_N
GND GND GND

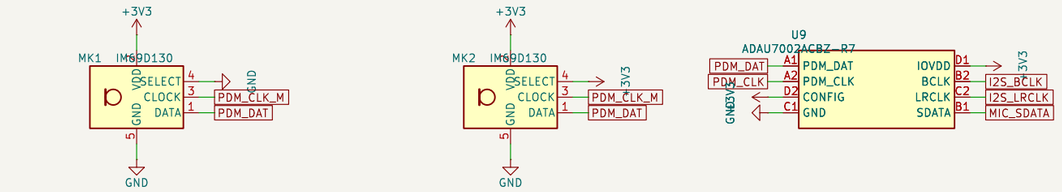
U6
TPS2553DBVR
IN
EN
GND
OUT
FAULT
ILIM
USB_PWR2 +5V GND
USB2_VBUS
USB_OC2_N
USB2_ILIM
R39 20k USB2_ILIM GND
R40 10k +3V3 USB_OC2_N
C40 4.7uF +5V GND

U8
USBLC6-2SC6
USB2_N
USB2_P
USB2_VBUS GND
C41 100uF USB2_VBUS GND
C42 100nF USB2_VBUS GND
PWR_FLAG USB2_VBUS

Generated by hardware/gen — do not edit by hand
VoxHub (open hardware)
Sheet: /USB 2.0 hub and USB-A host ports/
File: usb.kicad_sch
Title: USB 2.0 hub and USB-A host ports

Size: A3	Date: 2026-11-11	**Rev: A1**
KiCad E.D.A. 9.0.9		Id: 4/6

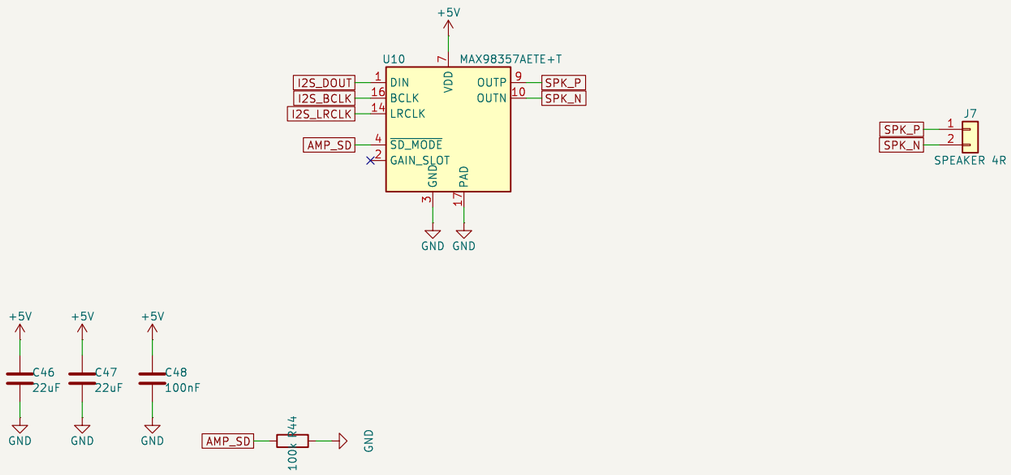
Stereo MEMS microphone array (bottom port through PCB) and PDM → I2S converter



MK1 SELECT=GND (left), MK2 SELECT=VDD (right); 70 mm apart

CONFIG = IOVDD → I2S output; BCLK = 64 fs, PDM clock = 64 fs (3.072 MHz @ 48 kHz)

Speaker amplifier (MAX98357A, I2S in, 3.2 W into 4 ohm from 5 V)



SD_MODE driven by GPIO16 (3.3 V = left channel, low = shutdown). GAIN_SLOT open = 9 dB.

Generated by hardware/gen — do not edit by hand

VoxHub (open hardware)

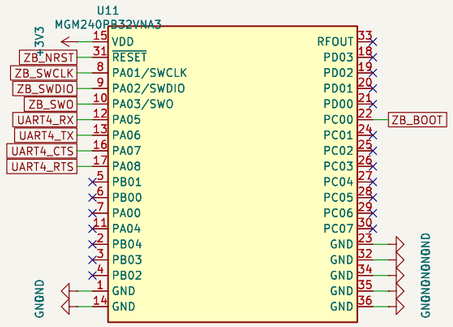
Sheet: /Audio: PDM microphones, PDM→I2S, class-D speaker amplifier/

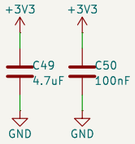
File: audio.kicad_sch

Title: Audio: PDM microphones, PDM→I2S, class-D speaker amplifier

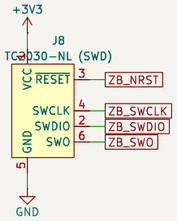
Size: A3	Date: 2026-11-11	**Rev: A1**
KiCad E.D.A. 9.0.9		Id: 5/6

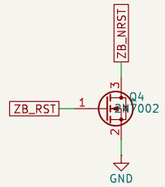
MGM240P network co-processor on CM4 UART4 (GPIO8–11, hardware flow control) + SWD













UART: PA05 = module TX -> GPIO9 (RXD4), PA06 = module RX <- GPIO8 (TXD4).
PA07 = module RTS -> GPIO10 (CTS4), PA08 = module CTS <- GPIO11 (RTS4) [datasheet fig. 5.1].
PC00 = bootloader entry, active low (GPIO25). Reset: GPIO24 HIGH pulls RESETn low (open drain).
SWD also wired to GPIO22/23 so the CM4 can flash the radio in-system (openocd bit-bang).

Generated by hardware/gen — do not edit by hand
VoxHub (open hardware)

Sheet: /Zigbee / Thread radio (MGM240P, EFR32MG24)/
File: zigbee.kicad_sch

Title: Zigbee / Thread radio (MGM240P, EFR32MG24)

Size: A3	Date: 2026-11-11	**Rev: A1**
KiCad E.D.A. 9.0.9		Id: 6/6